Ordinal measurement level has names and order.

Examples are:

The alphabet A B C D E F G H I J…

months in the year Jan, Feb, March, April, May…

days of the week Sunday, Monday, Tuesday…

All have names and order.

Can you think of more examples?

Ordinal numbers or numerals show relative order.

First, second, third… but they don’t give an idea of quantity. Saying that someone is second in line, you don’t know if they are also the last or if there are a million or more behind them.

For Ordinal levels of measurement, you can put them in order, but there is not a set interval between them.

Cardinal numbers have quantity and come in order. In the US, we use a base-ten number system and the number names also represent a specific amount, length, quantity, etc.

1 2 3

Counting numbers are the whole numbers that start at zero or one.

Mandarin and some other languages use explicit number naming, which makes it easier for the brain to process the numbers in the teens. Mathcounting is a project to help spread the idea of using explicit number naming in English and other languages that don’t use explicit number naming.

Why mathcounting?

Our English counting system has names for the teens that are very confusing! Old English was a base twelve system so eleven and twelve are from that. I have no idea where the mixed-up names for the other numbers came from.

Research and common sense both show that using explicit number names can make math thinking and learning easier.

The Mathcounting Project wants to:

(1) promote mathcounting and

(2) provide research opportunities to research the promotion, adoption, and science of mathcounting.

What is mathcounting?

1 one	11 ten-one	10 ten
2 two	12 ten-two	20 two-ten/twenty
3 three	13 ten-three	30 three-ten/thirty
4 four	14 ten-four	40 four-ten/forty
5 five	15 ten-five	50 five-ten/fifty
6 six	16 ten-six	60 six-ten/sixty
7 seven	17 ten-seven	70 seven-ten/seventy
8 eight	18 ten-eight	80 eight-ten/eighty
9 nine	19 ten-nine	90 nine-ten/ninety
10 ten	20 two-ten	100 one hundred is ten-ten

We use a place value system to write the numbers in base ten, with ten digits.

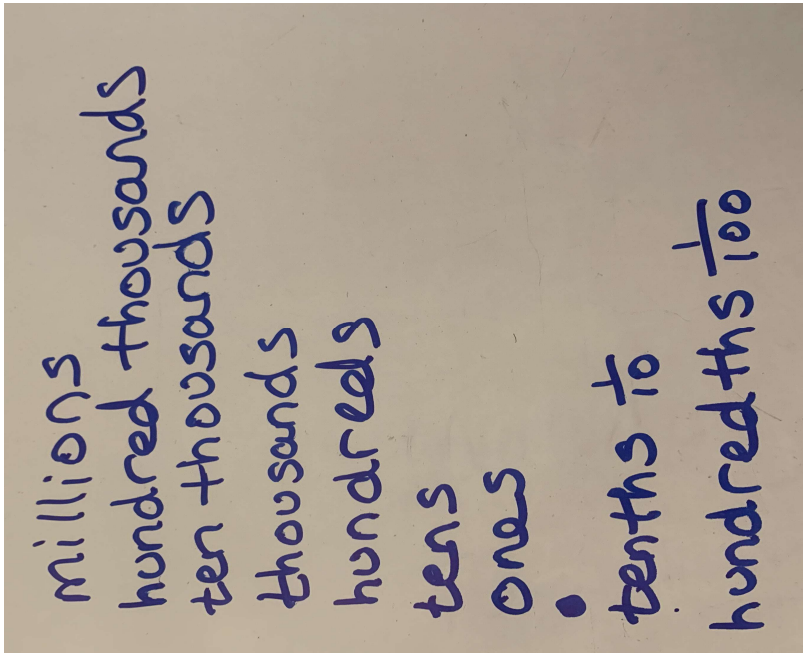
Since we say the largest numbers first, the numbers get higher in value to the left, because that is the direction that we write. Place value chart (math.net)

Ten thousands	Thousands	Hundreds	Tens	Ones		Tenths	Hundredths
8	7	4	8	1	.	2	3

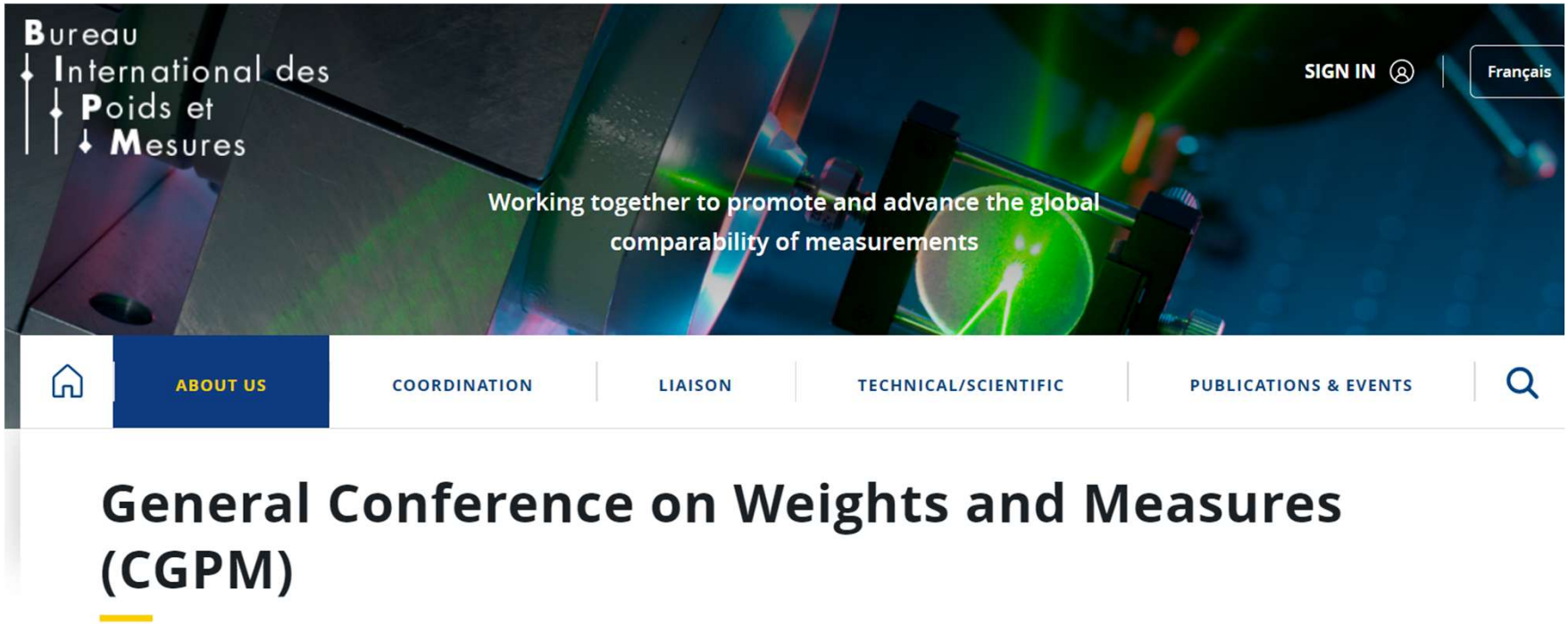
This tells us that 87481.23 has 8 ten thousands, 7 thousands, 4 hundreds, 8 tens, 1 one, 2 tenths, and 3 hundredths:

$$8(10000) + 7(1000) + 4(100) + 8(10) + 1(1) + 2(0.1) + 3(0.01) = 87481.23$$

You can practice with a place value chart to write numbers.



There are new names for bigger numbers! Scientists expand world's measuring unit systems for the first time this century - Los Angeles Times (latimes.com) cgpm – BIPM



With ordinal data, which has order, you can sort. When you line up data in order, you can look at the number of data points in a category that has order.

We can order the names in alphabetical order:

Aury, Birdie, Doc, Jae, Simba, and Socrates.

Since there is order, these are sets with ordering added, and we can use the term data set to be more inclusive. (As opposed to a “pure” set.)

1, 2, 2, 3, 4

Notice that we can’t call this data a mathematical set {1,2 ,3, 4} because elements are repeated.

We can use the term dataset Data set – Wikipedia or multiset (in Computer Science) for collections of data that can have repetitions.

I will use the term datapoints instead of elements, since we are talking about datasets now.

1, 2, 2, 3, 4

Sometimes square brackets are used for datasets or sets that can have duplicates. [1, 2, 2, 3, 4]

In computer science, a list is a finite (it has an end) collection of items or data that has order.

List (abstract data type) - Wikipedia

Ordered Pairs and Tuples

With a set, we didn't care about the order and there were no duplicates.

{1,2,3}

With a dataset or list or array, we can have duplicates with our datapoints. [1,2,2]

Tuple: a single item that has multiple components or pieces and the order of the pieces matters. (1,2) or (1,3,3) or (x,y,z)

Ordered pair: a single entity that has two ordered pieces like (1, 3) or (2, -5). In algebra we use (x,y) as our standard ordered pair.

Ordered triplets have three components (x,y,z)

Beyond that, they are called **n-tuples** with n elements.